

SHIVAM TRIPATHY

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SUMMARY

Robotics Software Engineer with expertise in autonomy, perception, and control systems. Experience building and deploying real-world robotic systems including UAVs, humanoids, and mobile manipulators. Strong background in SLAM, state estimation, motion planning.

EDUCATION

Carnegie Mellon University – The Robotics Institute M.S. Robotic Systems Development GPA: 3.87/4.0 <i>Commencement Speaker, School of Computer Science (2024) • Swartz Entrepreneurial Fellow (2022) • TA: Systems Engineering for Robotics (2024)</i> Selected Coursework: ML, Robot Planning, Computer Vision, Robot Learning, Visual Learning, SLAM	May 2024 Pittsburgh, PA
MIT World Peace University B.Tech Mechanical Engineering GPA: 8.75/10 <i>Special Achiever Award (2021)</i>	May 2021 Pune, India

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, PyTorch, ROS2, Gazebo, NVIDIA Isaac Sim, MoveIt/Nav2, PX4, MAVLink, Git, OpenVINS VIO
Hardware & Sensors: RGBD Cameras, LiDAR, IMU, GPS, NVIDIA Jetson, Raspberry Pi, EO/IR Sensors
Robotics & AI: SLAM, Visual-Inertial Odometry, Motion Planning (RRT*, MPC, LQR), Kalman/Particle Filters, Imitation Learning, Reinforcement Learning, LLM-based Task Planning, Computer Vision (YOLO, CLIP)

WORK EXPERIENCE

Apellix Robotics Software Engineer - Led autonomy stack development for power-wash and spray-painting UAVs deployed in commercial and US Army applications. - Integrated visual localization and SLAM for GPS-denied autonomous navigation on Apellix drone platforms. - Managed full flight software stack: state estimation, controls, state machine, collision avoidance, and failsafe systems. - Implemented vision-based end-to-end imitation learning framework mapping raw camera input to flight control outputs. - Built Ground Control Station app with autonomous draw-to-clean feature, enabling a SaaS business model.	Jun 2024 – Present
CMU Robot Caregiving Lab Graduate Research Student - Built full autonomy stack for an assistive mobile manipulator: LiDAR + stereo + IMU fusion for SLAM (AMCL), A*/DWA planning, and 3D obstacle avoidance. - Developed cloud-integrated LLM agentic framework connecting natural language user requests to HRI, navigation, and manipulation pipelines. - Successfully deployed robot in a Pittsburgh care facility; work featured by CMU School of Computer Science.	Sep 2022 – Dec 2023
Symbol Robotics Robotics Software Intern - First employee; built baseline robotics software stack for a 32-DoF humanoid mobile manipulator. - Implemented RTAB-Map Visual-Inertial SLAM with dual RealSense D435i cameras for localization; integrated ROS-Nav + MoveIt! for autonomous navigation and manipulation. - Devised long-horizon task planning agentic framework using GPT-3.5.	May 2023 – Aug 2023
Asteria Aerospace Robotics Systems Engineer - Developed target position estimation algorithm for surveillance UAVs fusing IMU, GPS, elevation maps, and YOLO detections. - Led 360° antenna tracking system and ground control software development for multi-rotor and VTOL platforms.	Jul 2021 – Jul 2022
DRDO / Sun Mobility Robotics Intern / Systems Intern - DRDO: Implemented PID control and sensor filtering for gait-assistance exoskeleton robots. - Sun Mobility: Built automation software for simultaneous multi-battery charging, achieving 1.5× faster charge times.	Oct 2020 – Jun 2021

SELECTED PROJECTS (CMU)

Assistive Robot Autonomy: Sensor-fused SLAM + LLM planning stack deployed in a real care facility for autonomous fetch-and-retrieval tasks.
Drone MPC: Applied Model Predictive Control on a simulated drone to track moving subjects while avoiding dynamic obstacles.
CubeSat State Estimation: Extended Kalman Filter for orbital determination of a 1U CubeSat in LEO; Particle Filter for differential-drive pose estimation.
Deep RL for Nuclear Fusion: Policy-based RL to maximize plasma pressure in simulated fusion reactors while avoiding plasma tearing instabilities.